Response Letter with Annotations

Citation (APA format, using Zotero or other bibliography software)

Brady, W. J., Wills, J. A., Jost, J. T., Tucker, J. A., & Van Bavel, J. J. (2017). Emotion shapes the diffusion of moralized content in social networks. *Proceedings of the National Academy of Sciences*, *114*(28), 7313-7318.

Annotations

1. **Purposes**
   a. Motivation (one-sentence summary)

The motivation of this study was to investigate the mechanisms underlying moral contagion in social networks, specifically whether moral, emotional, or moral–emotional language drives the spread of political discussions and how these dynamics differ within in-groups, out-groups, and between groups.

   b. Specific Rationale (several bullet points)

- Little has been studied how social networks transmit moral attitudes and norms
- Emotions tend to be highly associated with moral judgments
- Moral emotions may be tied specifically to behavior that is relevant to morality and politics
- Communications about morality and politics with social networks are computer-mediated
- Some topics may yield different results
- Moral emotions may spread within in-group demonstrating "echo-chambers"

   c. Research Questions and/or Hypotheses

Main H: The presence of moral emotions in texts likely increase the "moral contagion"
RQs: - Does moral contagion depend on moral–emotional language, or is it driven by emotion alone?

- Does moral contagion operate through negative emotions only, or through both positive and negative emotions?
- Do specific emotions differentially drive moral contagion?
- Does moral contagion spread primarily within political groups, or also across group boundaries?

● Identify IV, DV, mediator, moderator

IVs - Moral, emotional, moral-emotional language count
DVs - Diffusion of moral contagion
Moderators - Negativity bias, political group networks & emotional valence
Conditions: Political topics

● Identify specific relationships among variables

- Emotional valence (positive/negative) × (Moral–emotional language → Message diffusion)
- Specific emotion type (anger/disgust/sadness) × (Moral–emotional language → Message diffusion)

- Group membership (in-group vs out-group) × (Moral–emotional language → Message diffusion)

2. **Method**
   a. Research Design
      - Twitter API data. n=563,312 tweets collected. n=313,003 tweets analyzed
      - Textual analysis
      - Twitter data, observational - archival
      - Three political topics: Gun control, same sex marriage and climate change
      - Data collection 22-42 days in 2015
      - Focused on natural occurring social networks
      - Frequency of words. Moral words dictionary + Emotional words dictionary LIWC
      - Human validation of moral, emotional and moral-emotional words at initial stages
      - Tokenization of tweets

   b. Sample (representative sample?)
      Not representative of population. Selected users with ideological preference. Excluded "verified" and neutral users

   c. Measures
      IVs

      - Distinctly moral language
      - Distinctly emotional language
      - Moral–emotional language (overlapping dictionaries)

      Moderators

      - Emotional valence (positive vs. negative)
      - Discrete emotions (anger, disgust, sadness)
      - Group membership (in-group vs. out-group based on ideology). Political ideology inferred using follower network–based latent space modeling

      DV

      - Message diffusion, measured as retweet count

   d. Analysis
      Main effects regression model
      IRR (incident rate ratio) was reported
      Negative binomial regression to model overdispersed retweet counts
      Interaction models for moderation (e.g., group boundaries)
      Robustness checks:

      - Alternative model specifications
      - Bootstrapping
      - Excluding clustered users

- Sensitivity analyses excluding moderates and verified users

3. **Results and Conclusions**

The presence of moral-emotional words in messages increased their diffusion by a factor of 20% for each additional word

a. Answers to RQs and Hypotheses

- **(Moral–emotional language → diffusion): Supported.**
  Messages containing **moral–emotional** language diffuse more widely than those with moral or emotional language alone across all topics.
- **(Negativity bias vs. general moral contagion): Partially supported.**
  Moral contagion is not limited to negative emotion; positive moral–emotional language drives diffusion in same-sex marriage, while negative moral–emotional language is more influential for climate change and gun control.
- **(Discrete emotions): Partially supported.**
  High-arousal emotions (especially anger) enhance diffusion in negatively framed contexts, whereas low-arousal emotions (sadness) consistently reduce diffusion; disgust shows no consistent effect.
- **(Group boundaries): Supported.**
  Moral contagion is stronger within ideological in-groups than across out-groups, indicating boundary-constrained diffusion.
- **(Ideological asymmetry): Confirmed.**
  Moral–emotional contagion effects are generally stronger in conservative networks than in liberal networks, particularly for climate change.

b. Include Key Figures/Tables

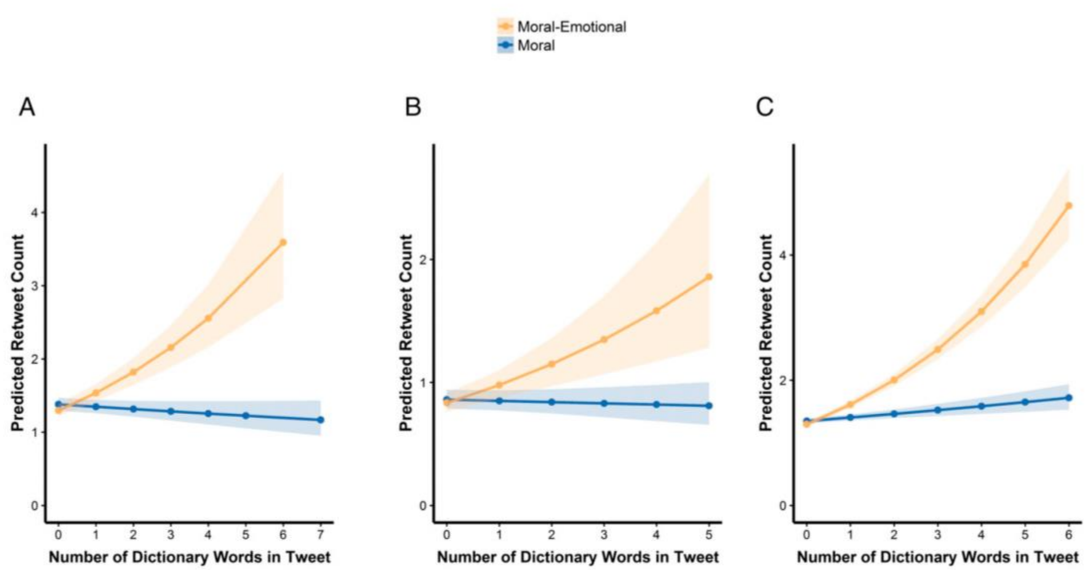

Fig. 1. Moral-emotional language predicts the greatest number of retweets. The graph depicts the number of retweets, at the mean level of continuous and effects-coded covariates, predicted for a given tweet as a function of moral and moral-emotional language present in the tweet. Bands reflect 95% CIs. An increase in moral-emotional language predicted large increases in retweet counts in the domain of (*A*) gun control, (*B*) same-sex marriage, and (*C*) climate change after adjusting for the effects of distinctly moral and distinctly emotional language and covariates.

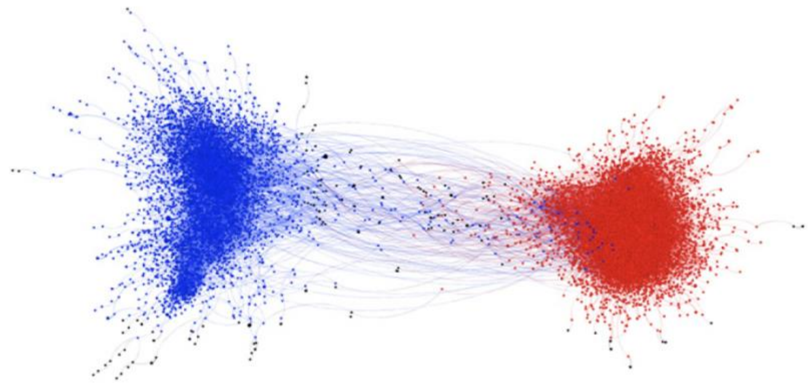

Fig. 3. Network graph of moral contagion shaded by political ideology. The graph represents a depiction of messages containing moral and emotional language, and their retweet activity, across all political topics (gun control, same-sex marriage, climate change). Nodes represent a user who sent a message, and edges (lines) represent a user retweeting another user. The two large communities were shaded based on the mean ideology of each respective community (blue represents a liberal mean, red represents a conservative mean).

c. Authors' Conclusions

The results of our studies also clarify how moral emotions contribute to moral contagion. One key finding was that morally framed emotional expressions explained statistical variance in diffusion of moral and political ideas over and above nonmoral emotional expression. We also observed that nonmoral emotions had a unique impact on social transmission for two out of three topics. Another key finding was that the expression of moral emotion aids diffusion within political in-group networks more than outgroup networks.Finally, our approach illustrates the utility of bringing a social network approach to bear on questions of moral psychology. In comparison with laboratory-based studies, the social network approach offers much greater ecological validity.

4. Discussion

a. Implications

- Extends moral psychology to network-level processes
- Helps explain political polarization and echo chambers
- Demonstrates the value of using dictionary based textual analysis for moral research

b. Limitations

- No causal inference due to the observational, correlational design.
- Measurement limitations from dictionary-based text analysis of moral emotions.
- Topic and platform specificity, with variable effect sizes and reliance on Twitter data.

c. Future Directions

- Experimental tests to establish causal effects of moral–emotional language on attitudes and behavior.
- Finer-grained moral emotion analysis, distinguishing subtypes of moral emotions.

- Network-structure and source effects, examining how influence, tie strength, and network density shape moral contagion.

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Response Letter (~600 words)

- Your assessment of the study
- Weaknesses
- Other notes

The study effectively used dictionaries to investigate moral contagion by examining which tweets are more likely to be retweeted depending on the presence of emotional, moral, or moral–emotional words across three politically polarizing topics. Although the study was not a true social network analysis that examined network structure or patterns, it focused on whether Twitter users retweet messages containing moral–emotional language from original tweeters who share the same political ideology. This approach allowed the authors to test their central claim that moral contagion is more likely to occur within ideological groups than across them. As the authors note, this creates opportunities for future work to examine network structures and the roles of specific users within and between groups.

A key strength of the study is its dictionary-based approach to identifying moral and emotional content. Tweets were first tokenized into individual words, with punctuation removed and user mentions excluded, and then matched against validated moral and emotion dictionaries. This step-by-step process makes the analysis transparent and easy to replicate. The distinction between distinctly moral, distinctly emotional, and moral–emotional words allows for examining how moral framing combined with emotion relates to diffusion compared to either component alone. As the authors argue that moral emotions are closely related to moral and political behavior, this makes a dictionary-based approach a reasonable way to capture moral emotions at scale.

Although the main manuscript is relatively short (6 pages), the Supporting Information (38 pages) substantially explains the paper by detailing how the dictionaries and tweets were validated. The authors explain how subsets of dictionary words were evaluated by trained coders and crowd-sourced participants to ensure that moral–emotional words were perceived as both moral and emotional. In addition, trained research assistants validated tweet relevance for each topic that the key-word approach sufficiently answers the research questions for the study.

At the same time, the dictionary-based approach also introduces limitations. While tokenization and keyword matching improve interpretability, they do not account for the broader context or intended meaning of tweets (Feuerriegel et al., 2025). For instance, sarcasm may not be fully captured by counting words alone. Again, the validation process focused on confirming that dictionary terms were morally or emotionally relevant in general, rather than training the model to learn meaning from the data. Still, as a keyword-based approach, this method prioritizes transparency and internal validity, which is an important strength in large-scale computational research.

Another limitation is that the study relies on observational data, which limits causal claims. Although moral–emotional language is associated with greater diffusion, it remains unclear whether such language directly causes increased sharing or whether highly shareable content is more likely to contain moral–emotional terms. The authors acknowledge this limitation and suggest experimental designs as a logical next step.

Finally, while the study tests ideological in-group versus out-group diffusion, it does not directly examine network structure or user-level influence. Retweets are treated as diffusion events, but factors such as centrality, tie strength, or the role of highly influential users are not modeled. This again highlights a natural extension of the study rather than a major weakness. Overall, the study provides useful insights into how moral–emotional language shapes the diffusion of polarizing topics online through dictionary-based analysis of large-scale social network data.

Reference:

Feuerriegel, S., Maarouf, A., Bär, D., Geissler, D., Schweisthal, J., Pröllochs, N., ... & Van Bavel, J. J. (2025). Using natural language processing to analyse text data in behavioural science. *Nature Reviews Psychology*, *4*(2), 96-111.